

MEIYARASU K

AI/ML ENGINEER | COMPUTER VISION | PYTHON DEVELOPER

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2 years of experience building production computer vision and LLM-driven systems. Shipped 2 production CV systems (ANPR, Face Recognition with anti-spoofing) at ~20 FPS real-time inference and curated 100K+ image pipelines from scratch, alongside self-driven projects in violence detection, dataset-annotation tooling, and LLM fine-tuning. Built an Anthropic Claude-powered natural-language-to-SQL reporting system across a 220+ table production database. Strong grounding in transformer architectures and FastAPI/AWS deployment.

TECHNICAL SKILLS

AI/ML Domains: Machine Learning, Deep Learning, Computer Vision, OCR, Biometric Authentication, Anti-Spoofing, NL-to-SQL, LLM Fine-Tuning, Instruction Fine-Tuning

Frameworks: PyTorch, OpenCV, Hugging Face Transformers, scikit-learn, NumPy, Pandas

Model Architectures: YOLOv8/v11, YOLO26 (s/m, P2 head), TrOCR, FaceNet, EfficientNet-B0, PaddleOCR, R3D-18

Backend & Cloud: FastAPI, AWS EC2, Docker, SQLAlchemy, MySQL, REST API Design, Anthropic Claude API, Streamlit, Ultralytics

Programming & Tools: Python, SQL, Linux, Git, openpyxl, ReportLab

KEY ACHIEVEMENTS

- Shipped 4 production/self-driven CV systems end-to-end — ANPR, Face Recognition + Anti-Spoofing, Stitch Counting, and Violence Detection.
- Curated 100K+ raw images (custom deduplication, blur-scoring, hard-negative mining) and manually point-annotated ~100K stitches for the counting model.
- Achieved 93% OCR accuracy (TrOCR), 81% / 0.81 F1 violence-detection accuracy (R3D-18), and ~20 FPS real-time inference across deployed systems.

PROFESSIONAL EXPERIENCE

AI/ML Engineer — Track Tech Solutions

Apr 2026 – Sep 2026

Tamil Nadu, India

- Built a stitch-counting CV system (FastAPI) using YOLO26s/YOLO26m with P2 heads — separate models for single-needle, double-needle, and overlock stitch types with dedicated scale-point detection; trained on a self-built dataset of ~100K manually point-annotated stitches.
- Implemented SAHI sliced inference for small/edge-stitch recall, class-agnostic duplicate removal (IoU + center-distance), and a scale-anchored band/border filter, plus a row-finder for double-needle/overlock counting and REST endpoints for single, annotated, and batch (up to 20 images) inference.
- Built a natural-language factory reporting chatbot (FastAPI + SQLAlchemy) translating plain-English questions into safe, read-only MySQL queries via the Anthropic Claude API across a 220+ table garment production database with almost no declared foreign keys.
- Designed a business-glossary/domain-group table-selection layer for reliable join-set selection, plus a SQL safety guard (SELECT-only, forbidden-keyword filtering, auto-LIMIT) and an empty-result investigator with fallback retries.
- Built a voice-to-database assistant using Claude API tool-calling with prompt-based filters to route transcribed voice queries — a hands-free extension of the reporting workflow.

Python Developer — Computer Vision — Hexbee Software Solutions Pvt Ltd

Sep 2024 – Apr 2026

Tamil Nadu, India

- Engineered 2 production CV systems (ANPR, Face Recognition + Anti-Spoofing) deployed live; curated 100K+ raw images with custom deduplication and annotation pipelines.
- Fine-tuned Microsoft TrOCR (93% accuracy) and built the ANPR pipeline (YOLOv8 + PaddleOCR/TrOCR, Indian-plate regex); two-round fine-tuning (14K detections + 10K hard negatives) enabled ~20 FPS real-time inference on AWS EC2/LAN cameras via FastAPI.
- Built a 3-stage face recognition pipeline (YOLOv11-face → EfficientNet-B0 liveness → FaceNet verification) trained on 10K self-collected anti-spoof images; deployed on AWS EC2.

KEY PROJECTS

- AI Camera Security — Violence Detection:** Fine-tuned R3D-18 (Kinetics-400) on RWF-2000 CCTV data, achieving 81% accuracy / 0.81 F1; batched GPU inference at ~20 FPS with a 3-tier Streamlit alert system.
- AI-Powered Image Annotation Tool:** Local-first CVAT-inspired tool (FastAPI + JS) with YOLO pre-annotation, human-in-the-loop review, multi-format export (YOLO/COCO/VOC/JSONL), and in-tool few-shot fine-tuning.
- Self-Persona AI (LoRA Fine-Tuned Career Chatbot):** Fine-tuned Llama-3.2-3B-Instruct via LoRA on 541 hand-built examples grounded in own project history, with assistant-turn loss masking and early stopping to prevent hallucination; dual-GPU torchrun training on Kaggle T4s, served via a 4-bit quantized Streamlit app.

EDUCATION

Bachelor of Computer Applications — J.K.K. Nataraja College of Arts & Science

Jun 2022

CGPA: 6.81 / 10

CERTIFICATIONS

Machine Learning with Python (IBM) • Data Visualization (IBM) • Data Analysis (IBM) • Deep Learning & Neural Networks with PyTorch (Udemy) • Python Data Structures & Algorithms (Udemy) • Complete Data Science, ML, DL & NLP Bootcamp (Udemy) • SQL Basic & Python Basic (HackerRank)